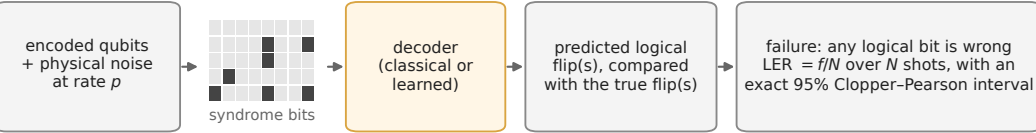


# The decoding task



## Three evaluation settings, each with its own reference decoder (never compared across settings)

| code                                                                         | noise model                                                  | decoder input                    | reference decoder               |
|------------------------------------------------------------------------------|--------------------------------------------------------------|----------------------------------|---------------------------------|
| <b>A</b> rotated surface code<br>$d = 5, 7, 9, 11$                           | circuit level: gates, measurements and resets can fail       | 25-round detector history        | MWPM (PyMatching)               |
| <b>B</b> bivariate-bicycle (BB) codes<br>[[72, 12, 6]], [[144, 12, 12]]      | code capacity: data flips only, perfect checks, one round    | one syndrome                     | BP-OSD (ldpc)                   |
| <b>C</b> BB codes [[72, 12, 6]] ( $T = 6$ ) and [[144, 12, 12]] ( $T = 12$ ) | phenomenological: data and measurement flips over $T$ rounds | $T+1$ rounds of detection events | BP-OSD on the space-time checks |